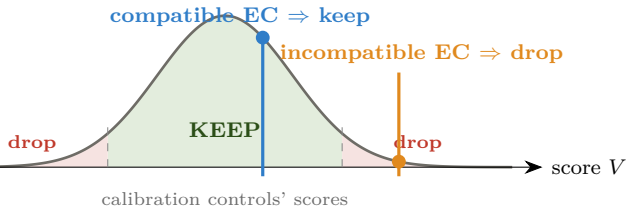


(1) fit model $\hat{m}(X)$
on a *training* fold

(2) on a held-out *calibration* fold,
score each control $V_i = \frac{Y_i - \hat{m}(X_i)}{\hat{s}(X_i)}$

(3) rank a new EC's V
among them $\Rightarrow p$ -value



Keep the MIDDLE band; drop *both* tails.

A score that is unusually *high* or unusually *low* is incompatible — so a p -value that is too small or too large is screened out.

Even if $\hat{m}$ is *misspecified*: controls and compatible ECs share the same score pattern, so the rank (hence the p -value) is still valid.